LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 9 Test (Set B): Light – Reflection and Refraction

Chapter / Topic: Light – Reflection and Refraction

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Use the New Cartesian Sign Convention. Show the formula, substitution and units in numericals. Draw ray diagrams neatly.

SECTION A

6 marks

Choose the correct option.

Q1. The radius of curvature of a convex mirror is 32 cm. Its focal length is: **[1]**

(a) −64 cm (b) −16 cm
(c) +16 cm (d) +32 cm

Q2. Which of the following is used as a magnifying glass? **[1]**

(a) Concave lens (b) Convex lens
(c) Plane glass sheet (d) Prism

Q3. Light travels fastest in: **[1]**

(a) glass (b) water
(c) diamond (d) vacuum

Q4. The magnification produced by a plane mirror is: **[1]**

(a) −1 (b) zero
(c) infinity (d) +1

Q5. When a ray of light passes obliquely from a denser medium into a rarer medium, it: **[1]**

(a) bends towards the normal (b) always goes straight
(c) bends away from the normal (d) is completely absorbed

Q6. **Assertion (A):** Convex mirrors are placed at sharp bends on hilly roads.
Reason (R): A convex mirror gives an erect image and a much wider field of view. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Define the power of a lens and give its SI unit. Find the power of a convex lens of focal length 40 cm. **[2]**

OR

What does it mean when we say that the refractive index of diamond is 2.42?

Q8. Which type of mirror is used (a) in solar furnaces, (b) as a shaving mirror? Give a reason for each. **[2]**

SECTION C

6 marks

Q9. An object 5 cm high is placed 20 cm in front of a concave mirror of focal length 10 cm. **[3]**

(i) Find the position of the image. **[2]**

(ii) Find the size and nature of the image. **[1]**

Q10. A convex lens of focal length 10 cm forms a real image three times the size of the object. Find the object distance and the image distance. **[3]**

OR

An object is placed 20 cm in front of a concave lens of focal length 20 cm. Find the position, nature and magnification of the image.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Draw a ray diagram for a concave mirror with the object placed between C and F. State the position, size and nature of the image. **[3]**

(ii) Two lenses of power +3.5 D and −2.5 D are placed in contact. Find the power and focal length of the combination. Is it converging or diverging? **[2]**

OR

(a) State the laws of reflection of light. [2]
(b) A coin at the bottom of a bucket of water appears to be raised. Explain why. [3]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. In an experiment with a rectangular glass slab, a student sent a ray of light into the slab at an angle of incidence of 30°. The angle of refraction inside the glass was measured as 19.5°. (Use sin 30° = 0.500 and sin 19.5° = 0.334.) **[4]**

(i) State Snell's law of refraction. **[1]**

(ii) Calculate the refractive index of the glass. **[1]**

(iii) If the angle of incidence is increased to 45°, what happens to the angle of refraction and to the refractive index? **[2]**

~ All the best! ~